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Green backlash and right-wing populism

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This Review delves into the politics of climate policy, specifically focusing on the so-called green backlash—that is, rising resistance from voters, parties and governments to the climate transition. We present a narrative review of the literature on the political consequences of climate policies, highlighting the presence of backlash among citizens negatively affected by decarbonization efforts. Populist right forces emerge as the primary beneficiaries of the backlash as they tend to be more sceptical regarding anthropogenic climate change and less supportive of climate policies. In turn, their electoral success has negative implications for countries' climate policymaking and performance. Finally, we draw insights from the literature to reflect on what can be done to improve the political sustainability of climate policies.

Responding to climate change requires effective policy action. In democracies, enacting climate policies requires broad public support and chiefly electoral backing for parties and candidates proposing ambitious climate action[1,2]. Understanding the politics of climate change is thus crucial to ensure successful mitigation action that is politically sustainable in the long term[3]. Recently, there have been challenges to popular and political support for the climate transition. A growing body of work on disaffection with climate policy stresses two key types of factors: (1) economic ones relating to the distributional consequences of climate policies themselves[1,4,5]; and (2) cultural ones connecting to a broader process of growing scepticism towards political and scientific elites[6–9]. On the economic side, climate policies, by generating 'winners' and 'losers' in the labour market and by imposing unevenly distributed costs on citizens, may trigger a backlash that rewards climate-sceptic parties and candidates, leading to the election of policymakers who are less sensitive to climate issues. On the cultural side, a shift away from parties backing climate policies may result from broader processes, resulting in reduced trust in politicians and scientific elites. Regardless of its drivers, a so-called 'green backlash' can undermine the political sustainability of climate transition over time and prevent the enactment of climate mitigation policies down the line. This backlash is likely to breed support for right-wing populist parties[10], a term used here as an umbrella category, encompassing radical and extreme right formations characterized by anti-immigration, nationalist and anti-elite rhetoric[11]. These parties have been very effective in channelling political discontent[12] and mobilizing it in opposition to the climate transition[13].

This narrative review reports on research dealing with the politics of climate policy along the conceptual map depicted in Fig. 1 (see Supplementary Information for details about the process of paper collection). Climate policies, through their economic and cultural repercussions, impact citizens' preferences regarding climate attitudes and voting behaviour. In turn, the shift in preferences impacts election outcomes. Finally, elected politicians influence the trajectory of future climate policies. Understanding these dynamics is key to improving the political sustainability of climate policies and deepening our analysis of the green backlash. Although the term 'green backlash' is widely used in public discourse, it lacks a consistent definition in the scientific literature. This Review defines the 'green backlash' as resistance by a substantial segment of actors within a political community (for example, voters, political parties and governments) towards the climate transition, which may manifest in efforts to resist, repeal or roll back climate policies[14,15].

The political consequences of climate policies

To investigate the political consequences of climate mitigation policies, we begin by considering studies that examine policies favouring renewable energy technologies, the main alternative to fossil fuel technologies. Then, we move to studies on policies that penalize fossil fuels. Finally, we review studies on the role of people's positioning in the labour market and how it might evolve as a result of climate policies.

Renewable energy policies might generate citizens' discontent due to their negative externalities, reducing electoral support for parties

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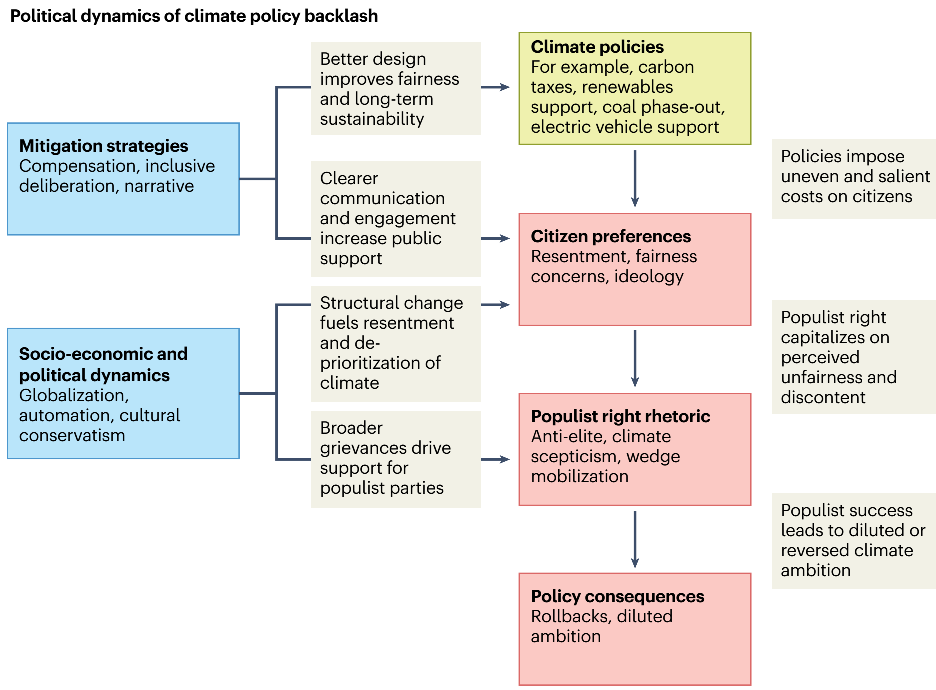

Fig. 1 | Structure and logic of the narrative review. Green and red boxes represent the steps of the process that are the object of the papers reviewed here. Blue boxes depict correction measures and external factors affecting the process flow. Grey areas describe the channels and detail the relationships.

responsible for their implementation. For instance, the installation of wind turbines creates negative externalities related to noise, impact on the landscape and effect on house prices[16]. This may lead to local opposition. An energy policy that liberalized the installation of wind turbines in the Canadian province of Ontario, by eliminating the veto power of local communities, led to substantial political backlash[16]. In electoral precincts with a proposed or operational turbine, the vote share for the governing party responsible for the policy dropped by 4–10% at the next election, compared with similar precincts without a turbine. This effect is estimated to account for around 6,050 lost votes by the incumbent party, and may have determined the election outcome, as the governing party lost its majority by just one seat. A moratorium on new wind turbines then followed to appease the protests of active anti-wind campaigners. A similar backlash has been observed with respect to wind turbines installed in Sweden[17], and with respect to both wind and solar developments in Germany[18,19]. That said, this electoral effect quickly fades with increasing distance from the renewable energy installations and is mitigated in areas where more tax revenues from energy generation accrue to local authorities[19]. Evidence in line with anti-incumbent effects of climate policy is also found at the macro level in a study covering 30 countries between 2001 and 2015[20].

Besides the anti-incumbent effect of renewables, studies also provide evidence for a polarizing effect. For instance, in German municipalities where wind turbines were installed, both support for the Green Party, the leading proponent of wind energy development, and support for the right-wing populist party Alternative für Deutschland (AfD), the primary opponent, increased[21]. This happened as policy supporters versus antagonists were increasingly mobilized in opposite directions in contexts where the issue of green energy became more salient. Conversely, focusing on regional elections in southern Italy, a study found that installing wind turbines generated a substantial and significant premium for pro-renewables left-wing regional administrations and a mild, not statistically significant electoral backlash against right-wing regional administrations[22].

At the broader electoral district level, the positive economic and environmental implications might win out against local resistance. Next to negative externalities, green energy projects also generate positive economic effects at the local level, for instance, in terms of job and firm creation or inflows of government subsidies. This could aid the incumbent or pro-renewable political forces. Such positive effects may not necessarily outweigh the negative externalities at the very local level, but this may be more likely the case as one considers larger geographic areas. In line with this intuition, a study investigating the political effects of wind turbine deployment in the USA, at the larger level of congressional districts, found that the installation of turbines increased the vote share for pro-renewables Democratic candidates[23]. In general, this study did not find evidence of a significant effect of turbines on support for the incumbents, neither positive nor negative. In contrast, another US study found that wind turbine developments positively impacted the electoral performance of incumbent parties, with Republicans benefiting as much as, or even more than, Democrats[24].

Next, we review studies examining the political consequences of policies penalizing fossil fuels. One such policy is traffic restrictions on fossil-fuelled mobility, which may generate public opposition among affected groups of citizens. For instance, a traffic ban on relatively more polluting cars in Milan generated a sizable political backlash benefiting the right-wing populist party Lega[4]. The populist right strongly opposed the policy, criticizing its regressive distributional consequences. This position was politically rewarding, as affected car owners were more likely to shift their support to Lega at the next election. Importantly, affected car owners who received some form of compensation from the municipality, such as a subsidy to buy a cleaner car or a public transport subscription, did not follow this trend. This result supports the idea that setting up compensation schemes can effectively enhance the public acceptability of climate policies.

Carbon taxes are a second key example of these policies. Like traffic bans, they can generate a political backlash through distributional consequences. Cross-country evidence shows that carbon taxes tend to be lower in political contexts characterized by higher electoral competition among parties[25]. A Dutch study investigated the political implications of a policy that increased household natural gas consumption taxes while redistributing the revenues as renewable

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energy subsidies[26]. The policy resulted in a 46% increase in gas prices, and around 20% of home owners installed rooftop solar panels. The policy had regressive implications and triggered fierce opposition from populist right parties, which obtained significant electoral gains among penalized groups. In particular, the backlash was particularly evident for individuals classified as 'energy poor', those who spend at least 10% of their household income on energy, struggle to pay their utility bills or are unable to heat their homes adequately. Along similar lines, a solar panel subsidization programme in Belgium has been found to reduce support for government parties while increasing votes for anti-establishment parties[27]. This result is mainly driven by non-adopters of solar panels, who are, on average, poorer than adopters and are subject to regressive electricity surcharges imposed to finance the subsidies. Concerning the interaction between material and cultural concerns, studies have found evidence of lower support for carbon taxes among individuals displaying lower trust in politics[28]. Interestingly, individuals with low political trust tend to be against carbon taxes even when they believe in anthropogenic climate change[29].

The third kind of policy aimed at penalizing the use of fossil fuels is related to the phasing out of coal. The move away from coal may generate political opposition in affected communities due to localized distributional consequences. In Germany, for example, closures of coal plants and mines between 2007 and 2022 were found to determine lower vote shares for the Social Democratic Party, a long-standing proponent of coal phasing-out, and increased abstention rates in affected municipalities[30]. In the Appalachian region of the USA, 54% of coal mining jobs (32,000 of 60,000) were lost between 2011 and 2016. A study at the county level showed that the loss of coal mining jobs increased the Republican vote share by approximately 4 percentage points in the presidential elections of 2012[31]. This effect persisted in 2016 and was equivalent to a vote shift of around three times the number of jobs lost. The approach to coal became politically very salient over this period, with the Democratic Party strongly favouring phasing out coal, and the Republican Party strongly opposing it. 'Bring back coal' was one of the key slogans of Donald Trump's 2016 presidential campaign, when he promised to cut back anti-coal regulations imposed by the Obama administration. Trump emphasized the tangible and immediate costs of the energy transition vis-à-vis potential future gains that may ultimately prove immaterial to those who reject the idea of human-driven deadly climate change altogether.

However, the political consequences of coal phasing-out may not be uniform across countries. A similar phasing-out process in Spain, for example, was accompanied by a Just Transition Agreement put forward by the governing Socialist Party shortly before the national elections of 2019. The agreement was negotiated with both the trade unions and companies involved, and entailed support for affected workers and investment plans for affected municipalities. Pointing to the effectiveness of this agreement, a study found that support for the Socialist Party increased by 1.8 percentage points in coal mining municipalities, compared with relatively similar non-coal-mining municipalities that were not subject to the agreement[32]. The study suggests that this result is probably due not only to the compensatory content of the agreement but also to the process by which it was designed and negotiated, which increased public awareness and support.

A final type of policy penalizing fossil fuels relates to the transition towards electric vehicles. Companies and workers specialized in producing internal combustion engines and related parts are vulnerable to this transition. The same applies, in turn, to geographic areas where employment in such production activities is concentrated, as the transition towards electric vehicles may lead to significant job losses. The actual, or feared, impact of this transition has the potential to become politically consequential. In the USA, the Republican Party gained a significant number of votes (2.5 percentage points) in counties more exposed to the electric vehicle transition in the presidential elections following the 2013–2016 period, when policies concerning electric vehicles became politically salient[33]. Along similar lines, another US study found that counties witnessing higher shares of employment related to the green economy displayed higher support for the Democrats since 2016[34]. This electoral shift was driven mainly by Republican incumbent districts with a high proportion of green employment that did not receive government support for green investment after 2016.

While the studies on coal phasing-out and electric vehicles speak of the importance of labour market concerns related to decarbonization policies for workers employed in polluting industries, the transition towards a decarbonized economy may also create labour market opportunities for workers that are complementary to the transition itself[35,36]. Perceptions of economic threats versus opportunities stemming from the transition may influence elections. Focusing on individual occupational profiles, which capture the whole range of employment prospects of individuals, rather than just their current occupation, a study found that individuals with greener occupational profiles vote more for environmentalist and green parties. In comparison, the opposite pattern emerges for people with browner profiles, who are also more likely to support radical right parties[37]. Focusing on the current occupation of individuals, workers employed in polluting industries or emission-intensive activities are found to be more likely to support radical right parties, less supportive of climate cooperation and stringent climate measures, and more sensitive to cost considerations when evaluating climate policies[38–40].

Taking stock of the evidence presented in this section, two main messages emerge. First, the distributional implications of climate policies tend to be politically consequential, generating a backlash among citizens who are negatively affected by these policies or fear being negatively affected in the future. Second, this backlash often benefits right-wing populist parties and candidates.

Populist right positioning on climate

We next investigate what makes populist right parties particularly attractive to voters and communities adversely affected by the decarbonization transition. To address this question, this section reviews the literature on populist right parties' positioning regarding climate change and climate policies, and how this fits their broader political strategy.

Several contributions to the literature document a consistent pattern of scepticism towards climate policies among populist right parties in western Europe. An analysis of their party manifestos shows that these parties strongly emphasize the trade-off between environmental protection and economic development; they are sceptical about man-made global warming, and overwhelmingly opposed to environmental taxes[41]. Indeed, populist radical right parties are the only party family to consistently express a negative stance towards climate protection, even as climate issues have become more prominent in their manifestos, particularly since 2019[42]. Other work shows that radical right parties adopt an oppositional climate policy rhetoric that diverges strongly from the mainstream consensus[13]. This is consistent with a political strategy aimed at politicizing 'wedge issues', that is, issues that cut across party lines, to split existing social coalitions. Survey data show that climate scepticism is widespread among voters across the political spectrum, so the mobilizing potential of opposition to climate policy is substantial. In this respect, populist right stances on climate change complement their strong positions on other wedge issues like immigration and European Union integration, as identified in earlier work[43].

A climate-sceptic stance aligns closely with the political narrative of the populist right. A defining feature of populist parties is their Manichean view of the world as a clash between 'pure people' and 'corrupt elites'[44]. Against this backdrop, right-wing populist rhetoric portrays climate change as a preoccupation of the urban elites and cosmopolitan climate activists, who design decarbonization policies

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that impose disproportionate burdens on ordinary citizens. As a result, these parties often emerge as the primary political opponents of climate policies, emphasizing their perceived unfairness and regressive implications.

Moreover, climate policy is increasingly identified with scientific expertise, technocratic management and the influence of multilateral institutions, frequent targets of right-wing populist resentment[4]. This association reinforces opposition to climate action, which is portrayed as part of a liberal, cosmopolitan elite agenda that stands in conflict with the nationalist and authoritarian values of populist movements[45]. Climate scientists and environmentalists are depicted as part of a corrupt elite driven by special interests and acting against the interests of the people. Moreover, the complexity of climate issues provides fertile ground for conspiracy theories, which tend to find fertile ground among populist right constituencies[46].

Looking beyond the general finding of populist right opposition to the green agenda, there is some differentiation in discursive patterns and rhetorical strategies adopted by specific parties. For instance, studies find that the German AfD tends to frame its opposition to climate change policy more in terms of 'response scepticism', that is, a criticism of the initiatives proposed and enacted to fight climate change, rather than outright denial of scientific findings[47,48]. Similar evidence is found for the French party Rassemblement National, which appears to be also somewhat less climate-sceptical than AfD in general[48]. As for the main Scandinavian populist right forces—that is, the Danish People's Party in Denmark, the Finns Party in Finland and the Sweden Democrats in Sweden— these parties have in common a strong criticism of what they call 'climate hysteria' and 'alarmism'[49]. Beyond this commonality, the Danish People's Party seems to have shifted over time from climate science denialism to 'climate policy conservativism', which entails acknowledging the existence of climate change but only supporting action when costs are minimal. The positions of the Sweden Democrats and the Finns Party are closer to 'climate policy nationalism', which entails arguing that climate change is due to actions of other countries and generally refusing calls for their own countries to act. This is close to response scepticism but incorporates a nationalist rationale as justification, emphasizing concerns that other countries may free-ride on domestic climate efforts. This said, all three parties share a more favourable attitude towards more traditional environmental issues, entailing national nature conservation and the protection of countryside landscapes.

A nationalist call for safeguarding a country's rural areas, the beauty and the integrity of the national environment, symbolically linked to the nation's culture, identity and pure people, is not uncommon among right-wing populist parties[45,50–52]. This approach emphasizes the right of Indigenous people to land and water resources, as well as the economic interests of rural communities. These communities would strongly benefit from effective climate change mitigation in the long run[53]. Yet, in the short to medium run, they would incur adjustment costs, for example, restrictions on some farming activities. Given these potential costs, opposition to climate action in the name of rural and agrarian interests may be politically rewarding. This is likely to be the case even in contexts where the share of employment in farming and fishing is actually low. This type of rhetoric has been shown to resonate with broader segments of the electorate, as citizens may sympathize more with farmers than with climate activists[54–57]. Finally, the nature conservation stance of right-wing populist parties also aligns with scepticism towards climate change mitigation measures that could impact landscape aesthetics, such as the installation of wind turbines.

The consequences of the green backlash

Having discussed the electoral consequences of climate policies and the positioning of populist right forces on climate issues, which makes them particularly likely to benefit from the green backlash, we now turn to the question of the extent to which this backlash is consequential for policymaking and environmental performance. Specifically, in this section, we review the available studies investigating what happens in countries where the influence of right-wing populist forces on legislatures and executives grows.

Arguably, the most evident climate policy effects of the electoral success of right-wing populists have been observed in the USA, under Donald Trump's first presidency. Several studies document and analyse in detail the various initiatives undertaken by the first Trump administration in this domain[58–62]. The withdrawal of the USA from the Paris Agreement, the cancellation of the Climate Action Plan initiated by the previous president, Barack Obama, the defunding of the Environmental Protection Agency, and the liberalization of oil and gas drilling even within national nature reserves were some of the most visible policy moves. They are consistent with Trump's scepticism over anthropogenic climate change and his aversion towards any measures that may, in his view, harm the economy and the broader American interest, especially when it comes to restrictions stemming from multilateral agreements. In this respect, Trump's approach constitutes a quintessential example of the populist right positioning on climate issues described in the previous section.

When focusing on Europe, the literature shows a consistent pattern of adverse impact of the strength of populist right parties on climate mitigation action. For instance, two cross-country studies find that greater influence of populist right parties on executives weakens climate action, specifically relating to greenhouse gas emissions reduction targets, renewable energy targets and phasing out of conventional energy production[63,64]. The impact is stronger when populist right parties directly control a relevant ministry, and mitigated when international commitments become more relevant, for instance, after the Paris Agreement of 2015. Other work finds that a greater presence of right-wing populist parties in executive institutions leads to higher greenhouse gas emissions per capita[65]. In contrast, the opposite holds for populist parties of the left, at least in southern Europe. A study documents how the climate-sceptical approach of populist right parties is not shared by populist parties of the left, which tend to display more pro-environment stances[64]. This is another illustration of the fact that, despite sharing populist rhetoric, left and right populist parties are fundamentally different when it comes to ideological stances and policy proposals in many domains[66,67].

Interestingly, in some cases, a greater presence of populist right parties in parliaments may positively impact climate action[63]. This happens if shrinking mainstream parties are forced to form broader government coalitions with smaller environmentalist parties, excluding the populist right. Importantly, research also suggests that the influence of populist right parties on climate action may vary across different political contexts[68]. There is evidence of more substantial negative effects of populist right influence on climate policy in majoritarian systems than in proportional systems. In fact, in majoritarian contexts, populist right forces may become dominant factions within mainstream political parties—a phenomenon known as 'majoritarian populism'—thus acquiring stronger policy influence. The trajectory of the US Republican Party under Trump is a case in point.

Overall, the evidence reviewed in this section suggests that, by increasing the prominence of right-wing populists within executives and legislatures, the green backlash could have significant implications for climate action.

Mechanisms and paths to address the backlash

We now turn to the question of how the green backlash can be addressed and what measures can enhance the political sustainability of climate policies. We delve deeper into three mechanisms that may link climate policies to shifts in voting behaviour among relevant segments of the electorate.

First, the green transition may invoke material concerns that affect political behaviour. Individuals negatively affected by the transition, due to worsening labour market outcomes and actual or feared costs implied by specific climate policies, are more likely to turn to populist right parties that oppose such policies by criticizing their distributional consequences. The decarbonization transition poses a significant challenge, as the economic and social costs of industrial restructuring, such as job losses and obsolescence of skills, often disproportionately affect workforce segments that do not directly or immediately benefit from these changes. A 'just transition' requires leading this transformation in a way that is as fair and inclusive as possible. To achieve this, it is essential to design transition strategies that minimize negative impacts on vulnerable workers and communities by ensuring access to necessary skills training, developing competences for emerging industries, and providing robust social protection measures[69]. Compensation for those who may experience acute costs or burdens of the climate transition, as well as broader policy approaches aimed at a just transition, may foster both the public acceptability of specific policies and the general political sustainability of climate action[4,32].

The second mechanism involves negative impacts of other structural economic changes, which may fuel negative attitudes towards climate change and climate policies[70]. The decarbonization transition is a dimension of structural change in the economy, akin to other facets of structural change, such as globalization and technological progress. All these phenomena tend to produce aggregate welfare gains but with 'winners' and 'losers', at least in relative terms. Losers of globalization and technological progress, having to deal with pressing economic grievances, may de-prioritize climate change and, therefore, reduce their support for environmentalist parties. For instance, a study shows that higher exposure to import competition leads to lower support for environmentalist parties and more sceptical attitudes about climate change[71]. Among the possible mechanisms, the study emphasizes how globalization losers, like those who may experience acute costs or burdens of the climate transition, are more likely to turn to populist right parties. This dynamic may lead voters, who are initially drawn to populist right parties for reasons unrelated to climate policy, such as economic distress linked to globalization or automation, to adopt the parties' climate-sceptic positions as a consequence of their political alignment. Research finds that people holding more populist and nationalist beliefs, and in general supporters of the populist right, are more likely to display climate change scepticism, distrust of climate experts and lower support for climate policies[72–75].Tackling the rising inequalities driven by structural changes that breed support for the populist right becomes essential to increase public support for climate action.

Finally, from a cultural perspective, research across several countries shows a consistent link between anti-establishment sentiment, low institutional trust and opposition to climate policies. For populist supporters, diminished trust in political and scientific institutions becomes an important channel through which people-centrist and anti-elitist beliefs fuel positions against climate action. Studies from Canada, Norway, Austria and the Netherlands find that lower trust in institutions, including political and environmental bodies, is associated with climate change scepticism and resistance to climate policies[76–79]. In this respect, waning support for climate action may be part of a broader process of rising scepticism towards the establishment. Hence, the sustainability of climate policies also hinges on efforts to increase trust in politicians and institutions as well as science, for instance, by investing in civic and science education.

Overall, ensuring the political sustainability of climate action requires not only mitigating the distributional consequences of climate policies themselves, but also addressing the broader inequalities driven by wider processes of socio-economic change[71]. Ultimately, the fight against climate change is also a fight for a more inclusive society. Inclusive policies may contribute not only by addressing material grievances but also by fostering greater trust in institutions. As discussed, the drivers of the green backlash extend beyond economic concerns to include cultural factors.

Conclusion

Political support for climate action is essential for a successful decarbonization transition.

Drawing on the literature, we have shown that climate policies can trigger a backlash by creating perceived 'winners' and 'losers'. Although the distributive impacts of climate policies are often smaller than those of climate change itself, which is likely to generate even greater and more unequal harms[80], they tend to be far more politically salient. Rising inequalities owing to other dimensions of structural economic change and broader cultural shifts, entailing declining trust in elites and institutions, may pose additional challenges. Prioritizing the mitigation of distributional impacts from climate change and climate policies is essential. Equally important is adopting compensatory mechanisms and embedding participatory processes in the design of climate policy portfolios. In addition to these practical measures, it is critical to reshape the narrative surrounding climate policies, emphasizing their potential to promote fairness and shared benefits in people's perception[81].

This Review mainly focused on European and North American countries at the forefront of experimenting with climate policies. As many other national policies are rolled out globally, possibly incorporating some of the lessons learned so far, future research should uncover whether and in what form a green backlash emerges in other geographical and political contexts. A different stream of literature is investigating the political implications of climate adaptation and disaster relief policies, which are characterized by concentrated benefits and diffuse public costs[82]. This research will offer valuable insights into the public acceptability of such policies and their potential interactions with mitigation efforts, as adaptation is expected to play an increasingly prominent role in the climate policy portfolio.

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G.M. conducted the literature search and identified the papers included in the Review. V.B., I.C. and C.E.D.V. conceived the idea for the Review and jointly wrote the manuscript. All authors discussed and approved the final version.

Competing interests

The authors declare no competing interests.

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